Two-Regime Models for Daily Soil-Moisture Estimation in Washington

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SUMMER RESEARCH SHOWCASE · 2026

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Project Overview

Why daily estimates?

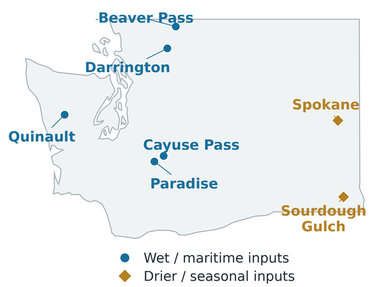
Soil moisture informs irrigation and drought monitoring. Ground sensors provide local measurements; satellites cover larger areas but leave gaps in daily observations.

We combine satellite, weather, and site data for daily soil-moisture estimation. Earlier work established the pipeline and a single global predictor.

THE SUMMER QUESTION

Can two specialists improve estimates across Washington's contrasting conditions?

Study Area & Data



Station colors show fitted memberships across western Washington, the Cascades, and the east.

7	54	5 cm
public stations	shared features	target depth

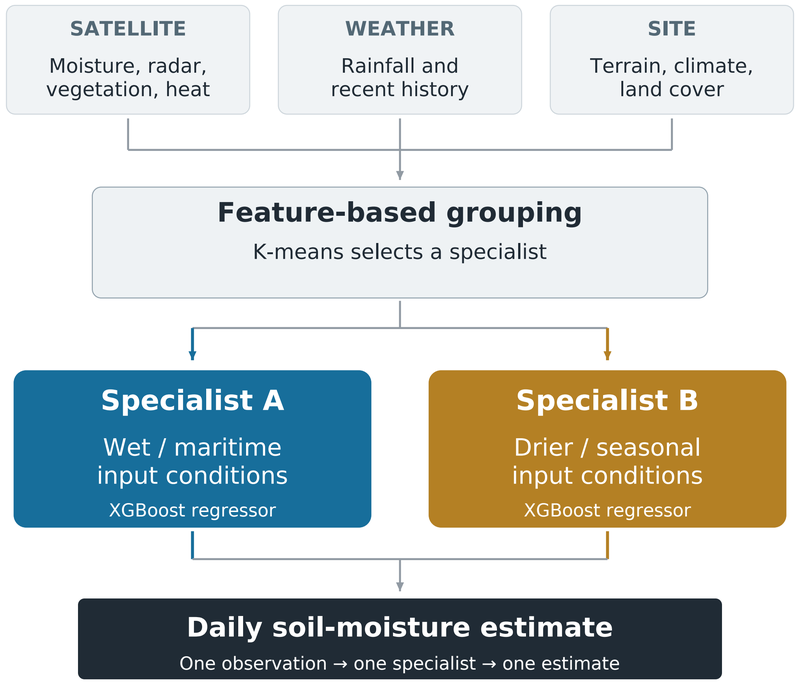
Satellite, weather, and site features are aligned to daily observations; public in-situ sensors supply training targets.

2017–2022 Fit period → **2023–2025 Later-year test**

Target: daily soil moisture (m³/m³).

Two-Regime Method

Group observations by their inputs, then fit an XGBoost specialist to each group's conditions.



Known stations keep one specialist; new stations are assigned using their features.

Grouping uses no observed in-situ targets.

What the Regimes Capture

Wet / maritime

5 training stations

Western Washington: coastal and Cascade conditions together.

Wettest-month rain: 391 mm
Satellite moisture: 0.414 m³/m³

Drier / seasonal

2 training stations

Eastern Washington: less rain and lower satellite moisture.

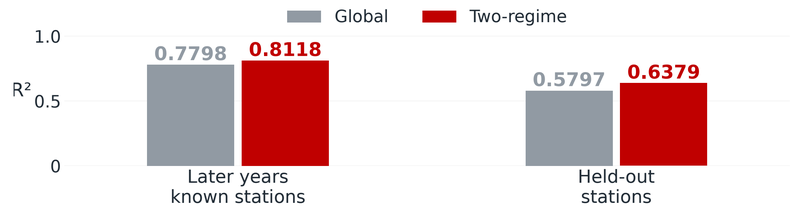
Wettest-month rain: 70 mm
Satellite moisture: 0.192 m³/m³

Group medians describe this Washington sample. Choose regime counts using local data.

Results

+0.0320 R² Later years

+0.0581 R² Held-out stations



Matched inputs and settings: 30 later-year runs; five per held-out station. Two-regime R² leads in every later-year run.

Where does specialization help?

Gains at four stations; losses at three. Eastern holdouts reduce representation of the smaller regime.

New Sensor Sites: Routing Matters

Missing SMAP inputs triggered automatic routing based on available precipitation history at five new ECE sites.

150 observations, July–August 2026; one fallback specialist. Full-season evaluation remains.

Takeaways

Specialization helps when routing is reliable and training sites represent new conditions. Next: evaluate more stations and seasons with the ECE team's sensors.

Sources: [1] Balkovec et al., IEEE AIIoT (2026), doi:10.1109/AIIoT68874.2026.11569136. [2] MDR Project, technical report 1.5 (2026).
In-situ data: NOAA USCRN and USDA ISMN SNOTEL. New sensor-site evaluation in collaboration with the Seattle University ECE team.